A Proposed Model to Transliterate Digitalized Braille Text into Gujarati: An Aid for Visually Challenged People

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Abstract. Information and Communication Technology (ICT), in the modern digital era has become so cutting-edge and touched to each and every field but there is still a disconnect between visually impaired persons and communication technologies because of ignorance and inattention. Due to the limited availability of resources and tools; people with visual impairments are unable to communicate in writing. In this paper, researcher has tried to draw the attention towards the challenges that are faced by visually impaired individuals in written communication and also discussed the benefits of having a tool that eradicate the communication gap. To overcome the challenges faced by blind people; author has proposed a model based on Natural Language Processing (NLP) and character mapping technique that act as an aid to transliterate the Braille text written by blind people to our regional language Gujarati. The technique used in the model uses standard Braille syntax. Rules are formed, lookup table is created, Braille characters are identified and they are transliterated to Gujarati language text. The model shows satisfactory result in the transliteration of Braille characters to Gujarati. It will also help visually impaired students in their education.

Keywords: Digital text processing · Braille text · Gujarati language · Transliteration · Natural Language Processing (NLP) · Character mapping · visually impaired people

1 Introduction

Language is essential to our everyday interactions because it allows us to communicate and express our feelings. It may be expressed orally or in writing [1]. Braille is a crucial written language for those who are blind or visually challenged [2, 3].

People with visual impairments are also valuable to the society who can contribute significantly to its prosperity [4, 5]. Consequently, it has become imperative to give them the tools and channels necessary to communicate with the outside world. Because of technological advancements; people who are blind or visually challenged can now access textual literature [5]. Through the use of Braille text transliteration, individuals who are visually impaired will be able to produce texts and converse with the sighted using a standard text format and comprehensible language.

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1.1 Braille Script

Braille [6] writing system is used by individuals who have low vision or are blind to read text, which is read with the tip of fingers and it consists of raised/embossed dots. Frenchman Louis Braille, the inventor of Braille; in order to improve night writing, developed code for the French alphabet [7–9]. In 1829, he published his technique, which included notation for music [10]. January 4th of every year is observed as an International World Braille Day to honor the creator of Braille. Using embossing, Braille is written using the notion of a cell—a thick sheet of paper with raised dots—on top of it. Six dots are placed in a Braille cell in a rectangular grid that consist of two horizontal dots in a row and three vertical dots in a column. Only sixty-four distinct patterns of dots may be generated as a result of this arrangement of six dots. A cell is any configuration of dots that contains a minimum of one embossed dot and a maximum of six embossed dots [1, 7, 8, 11] as shown in Fig. 1 and Fig. 2 represents printed Braille sheet.

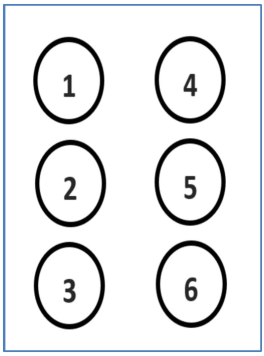


Fig. 1. Braille cell with six dots used to form Braille character [12]

Typically, a Braille printed sheet as shown in Fig. 2 has forty cells in each row and maximum of twenty-five rows of text on a single page. A typical Braille page measures around eleven inches by eleven inches. Although the Braille cell's dimensions are defined, they may differ significantly depending on the country's preferred script. A Braille script is also written as a cell consisting of eight dots, depending on the language that needs to be represented in terms of Braille. A dimension of Braille cell as printed on an embosser [2, 14, 15] is shown in Fig. 3.

Braille printer also known as embosser as shown in Fig. 4 is used to print/emboss Braille characters on the thick sheet exclusively meant to print Braille. The dots on the Braille sheet falls on the intersections of an orthogonal grid. Braille can be printed one-sided as well as two-sided. If it is printed two-sided in terms of recto and verso dots; the pattern of verso dots is somewhat shifts between recto dots so that it does not overlap each other as shown in Fig. 5.

The Braille script known as Bharati Braille (Indian Braille), also called Bhartiya Braille, is primarily used to write the languages of India. After India attained independence, for different languages and in a different region of the nation, there were eleven

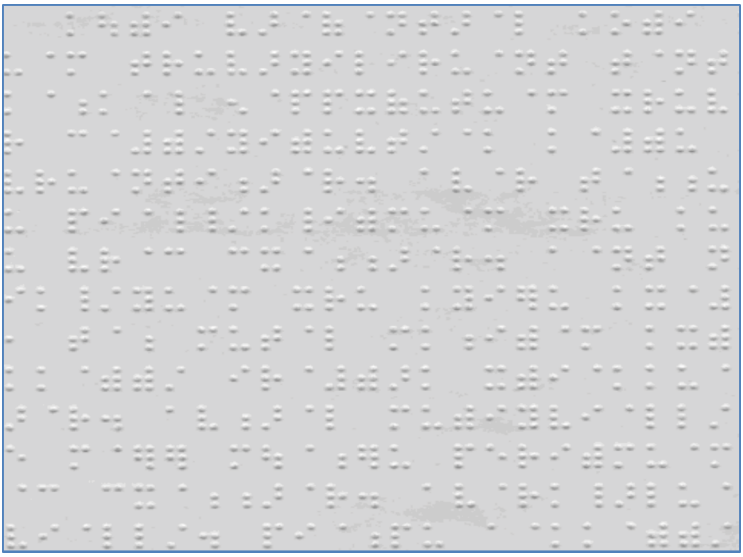


Fig. 2. Printed Braille sheet (Embossed Braille cells) [13]

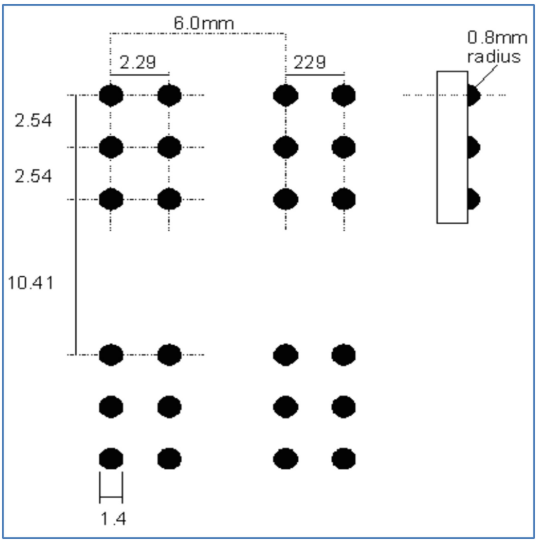


Fig. 3. Dimension of Braille cell carrying six dots [16]

Braille scripts in use [5]. Gujarati braille is comparable to Hindi Braille is also one of the Bharati Braille languages.

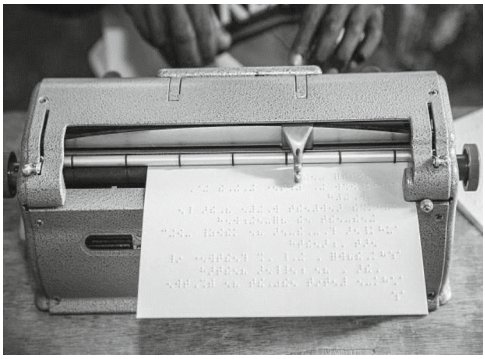


Fig. 4. Braille Printer/Embosser used to print Braille cells

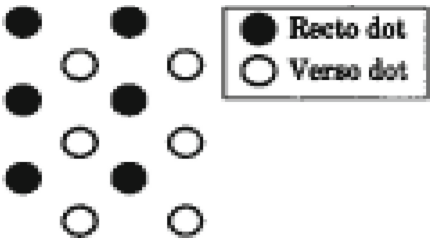


Fig. 5. Position of Verso and Recto Braille dots in two-sided print [15]

1.2 Gujarati Language

Gujarati is a regional language of Gujarat state and one of the scripts used in India for both oral and written communication. Gujarati script is multifaceted [15]. Gujarati characters are written from left to right and each character represents a syllable [17]. Consonants and vowels are referred to as Vyanjan and Swar respectively. Gujarati language uses a system of Maatras; a unique modifier symbols, one for each vowel, to modify the sound of consonants [2]. It has total 75 distinct characters [15]. Gujarati numerals and punctuations with their Braille representation are shown in Fig. 6 and Fig. 7 respectively. Figure 8 represents Gujarati characters with its Braille representation.

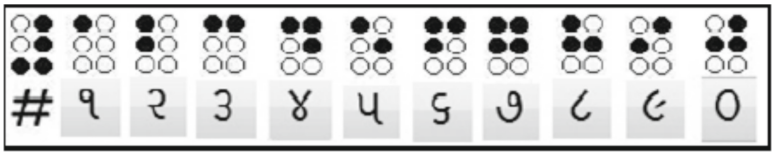


Fig. 6. Gujarati numerals and its corresponding Braille representation [4]

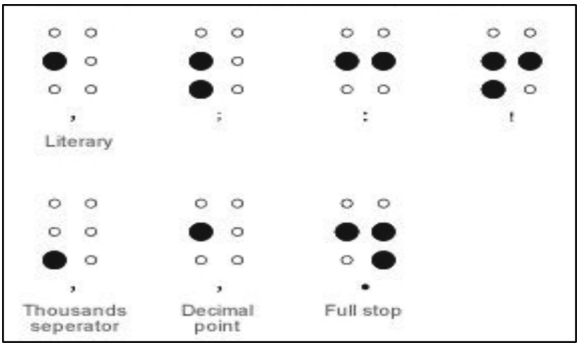


Fig. 7. Punctuation marks in Braille

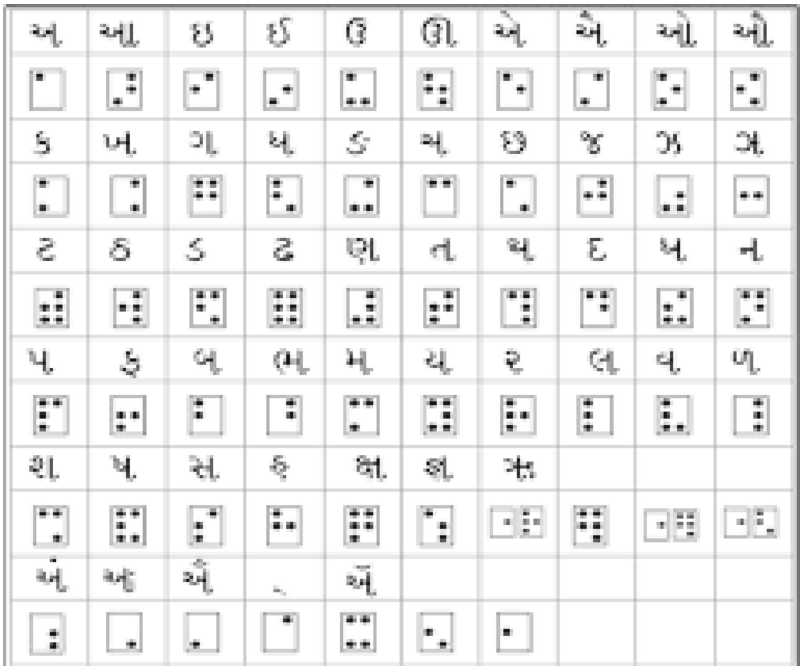


Fig. 8. Gujarati and its corresponding Braille representation [4]

1.3 Challenges of Visually Impaired People and Benefits of the Proposed Model

The primary problem is to generate Gujarati text from Braille. So, the visually impaired people can also communicate in written form with the sighted people and they don't remain dependent on anyone for conveying information in written form to the society.

If blind person wants to type some document, then the person has to be dependent on the people who will type for them in Gujarati language. But through proposed model, visually challenged people will be able to type their document in Braille and it will be

converted to Gujarati. So, they will not need help of others and sighted person will able to read.

Similarly, if visually impaired person wants to set the paper and type it, then again, the proposed model will be useful. As they will be able to type in Braille and through the system it will be converted to normal Gujarati language.

If the proposed model is developed then unemployed visually impaired person will be able to get some job such as data entry and they can perform data entry in Braille and then it will be converted into Gujarati language.

Similarly, it will also be helpful to the visually impaired person who is capable of getting job as PO or clerk, but due to their blindness they are not getting this opportunity.

It will also be helpful to the parents who are blind but their child is well sighted. They will be able to help their child in studying. For example, if child ask their parents that how to write some word in Gujarati, then the parents will able to write in Braille and their child will not be able to understand. But through the proposed model, if parents write some word in Braille, then it would be easily converted to Gujarati language which is understandable to their child.

It will also be useful in academic field. For example, if visually impaired person is teaching Gujarati language and that person want their student to write something then he has to either dictate or ask some other person to write it on the board. But now most of the schools have projectors attached with computer for teaching. So, if the teacher wants to give notes to the students, then teacher will write notes in Braille and through the proposed model it will be converted to normal Gujarati language.

If visually challenged person wants to type the content of the book, then also it will be possible with the help of proposed model.

So, in short, if the proposed model is commenced then it will be beneficial to visually impaired people in many ways by getting jobs in various sectors such as bank, corporate, schools etc. and it will reduce the written communication gap between sighted people and visually impaired people.

1.4 Need of the Research

In India there are thousands of languages, so there isn't any standard script used across the languages. For visually impaired people, six-dots Braille is a standard and unique script.

The National Institute for the Empowerment of Persons with Visual Disabilities (NIEPVD), then known as the National Institute for the Visually Handicapped, was also profoundly involved in the tuning of Bharati Braille which is meant for Indian languages. For the most of the official languages of India, NIEPVD Braille development unit has contributed notation systems for music, Science and math as well as shorthand system, Braille contractions and abbreviations [13].

Currently, there are 70 million vision impaired people and 4.95 million (0.36% of the population) blind people in India, out of which blind children are around 0.24 million [18].

In 31 districts in India; Community Ophthalmology, R P Centre, AIIMS conducted NPCB&VI National Blindness survey, among people aged 50 years. From Gujarat Kheda district was considered as a part of this survey. Prevalence of blindness was 1.15% which

was less than national average 1.99%. Prevalence of visual impairment was 11.13%. Prevalence of functional low vision (FLV) in district Kheda, Gujarat was 1.4% [18].

India - the second most populated nation contribute one-third to the world's blind population. But in India the Braille literacy rate is far lower (only 1%) than the regular literacy rate which is 77.7% [19] as shown in Fig. 9. Braille education will help visually impaired children and adults to increase their literacy. In India, audio content and Braille books are access by just 6.86% of schools [20].

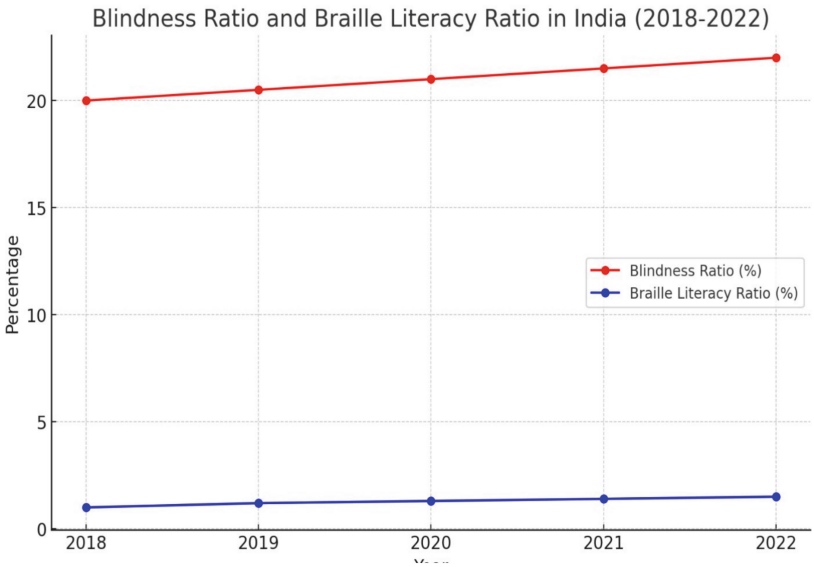


Fig. 9. Comparison between blindness and Braille literary ratio [19]

2 Literation Review

Following research work are carried out for Braille to text transliteration.

With reasonable constraints, J. Mennens, G. Francois, L. V. Tichelen and J. J. Engelen have created a method which translates Braille content into a format that can be read by computers. Therefore, an electronic Braille embosser can be used by a Braille printing business to recreate text that is stored on a computer [21].

P. Blenkhorn talked about how to convert Braille, which is saved as characters in a computer, into print. To do so, it mostly used a table-driven approach. Numerous character sets and languages are supported by the way the system is constructed and changeable. The algorithm is explained by taking into account the printing of Standard English Braille. Also, the tables used for the conversion are provided by the author [8].

A. S. Al-Salman describe a bi-lingual, bi-directional, editor-equipped translation system that is Arabic-Braille. It does not require any pricey machinery. The system can be expanded to support literate people's communication across languages, geographic

regions, and disabilities (visually impaired or sighted) if the necessary rule file is available for any other languages [6].

P. Rajarapollu, D. Laghate, S. Kodolikar and A. Khavle demonstrate how to use the Field Programmable Gate Array (FPGA) Spartan3 kit to convert Braille to text or speech. The genuine Braille language is translated into English in the regular domain. The Braille keypad, which has various cell combinations, is used to enter data into the system. The Spartan3 kit of FPGA then receives the additional input. Based on the combinations produced, the FPGA uses decoding logic in VHDL (VHSIC (Very High Speed Integrated Circuit) Hardware Description Language) to translate the Braille text into the appropriate English language text. The algorithm decodes the corresponding alphabet, converts it to speech, and displays it through the LCD interface with the Spartan3 kit [22].

G. E. Legge, J. S. Mansfield and C. M. Madison describe a brand-new, uniform technique for gauging Braille reading speed that was developed from the MNREAD print reading speed test. One of the main principles is to utilize content that is well-characterized and made up of basic terminology. The text should be provided in a standard spatial structure, and the sample length should be expressed in characters. 44 seasoned individuals are studied for Braille reading speed using the MNREAD test. A person reading 7.5 characters per second, or 124 words per minute, was the median speed. The reading speed of Grade 1 was 71.5% of Grade 2 when expressed in words per minute. This result supports the idea that the best criterion for assessing Braille reading speed is characters per second. A comparison between reading speed in print and Braille reveals that reading speed characteristics are both qualitatively and quantitatively equivalent when viewing conditions and measurement techniques are matched [1].

A. S. Al-Salman describes a translation/editor system for bi-directional and bi-lingual Arabic Braille that does not need expensive equipment. With proper rules for other languages and regardless of the disabilities of the people whether they are visually impaired or sighted, geographical locations or languages; this system can be generalized to facilitate communication among literate people [6].

A. Chaudhary, P. Garg and A. Agarwal present a new algorithm using image processing is developed to remove misalignment of scanned Braille pattern. It is known that this error in the scanned Braille document is occurred because of scanner imperfections. To fix the corner pixel value of tilted Braille cell pattern, the paper represents indexing method and it employs basic triangular property. So, they have calculated the angle by which the corner dots have been thrown out of alignment on a respective side. Sorting method is used for symmetrical and unsymmetrical Braille pattern, as a variety of different Braille patterns are available where some are regular and exactly symmetrical. The system is developed in MATLAB (R 2010a) and removes misalignment of Braille pattern and yields almost perfect results with acceptable errors [9].

H. Vyas and P. Virpariya has discussed template-based transliteration of Braille text. Author has used image processing to recognized Braille characters and then converted into text. The model is developed using python and MySQL. They have also discussed methodology in detail with template matching technique [23].

S. Shokat, K. Khan, R. Riaz, S. Rizvi, F. Rian, S. Kwon has discussed the comprehensive literature review on various Braille input methods and their benefits to visually impaired people. Authors discussed that blind people have difficulty in accessing the

technologies. The paper emphasis on the need of the system which reduces cognitive load on visually challenged individuals. The authors identify the research gap in this area as very less amount of work is done outside USA and Canada. So, they suggested that if Braille input system is developed it will be beneficial to the blind people [24].

U. Beg, K. Parvati, V. Jha have identified that there is a major gap in the available literature related to text to Braille conversion. Author mentioned that to enhance the communication for visually impaired people conversion into Braille will act as an aid and it will also reduce the illiteracy rates. Researchers pointed out that real-time applications such as instant text-to Braille and Braille to text could improve the accessibility for blind people. They have discussed that image processing technique can also be used to convert optical text to Braille [3].

S. Padmavathi, S. Sphoorthy Reddy, K.S.S. Manojna, D. Meenakshy has discussed the concept of Braille translators which converts written text into Brille cells. Researchers have also proposed the method to convert scanned Braille document into English, Hindi and Tamil text. They have also mentioned that this text can further be converted into the spoken form with the help of Speech Synthesizer [25].

V. Jha and K. Parvathi highlights the challenges faced by visually impaired people of society in accessing educational literature in Hindi. They also reviewed existing techniques for Braille translation as well as introduced machine learning approach that recognize and transliterates characters. With the help of their research education system and visually challenged people will be beneficial [26].

C. Li and W. Yan have described significance of Braille for blind people which allows them to read and write. They have discussed deep learning based efficient Braille translator tool as well as word-based Braille detection using ABCNet. They highlighted the importance of the tool as it will improve communication for blind people using advanced technologies [27].

T. Shreekanth and V. Udayashankara have emphasized on the need of Optical Braille Character Recognition (OBR) to store and preserve Braille documents and improve the communication between visually impaired people of the society and the sighted people. They have tried to recognize the Braille cells which are printed two-sided where verso and recto dots are present in the printed Braille document. They have worked on Hindi language Braille character recognition [28].

M. Dhanalakshmi and K. P. S. G. Sugirtha describes a new system to identify the Braille cells in six-dots from the scanned document and transliterate it into English language. Image processing phases such as preprocessing and segmentation are applied to the scanned document to recognize the Braille character and Euclidean distance is calculated. Further lookup table is considered to transliterate the Braille character to the corresponding text [29].

We found that there are many literatures in various languages that convert Braille script text to plain language text including Gujarati language. Researchers have also worked on Optical Character Recognition of Braille cell. But due to the complexities and syntax of Braille script, incomplete solutions are available and along with that all the tactics have their own limitations. So, to resolved the problem authors have tried to proposed transliteration model that uses text processing method, transliteration technique

and character mapping algorithm to convert Braille characters to plain Gujarati text. Let's look into it in detail.

3 Proposed Transliteration Model

The proposed transliteration model has several phases that process the Braille cell and transliterate it into a Gujarati character. Following section describes and represents the phases of the model as shown in Fig. 10.

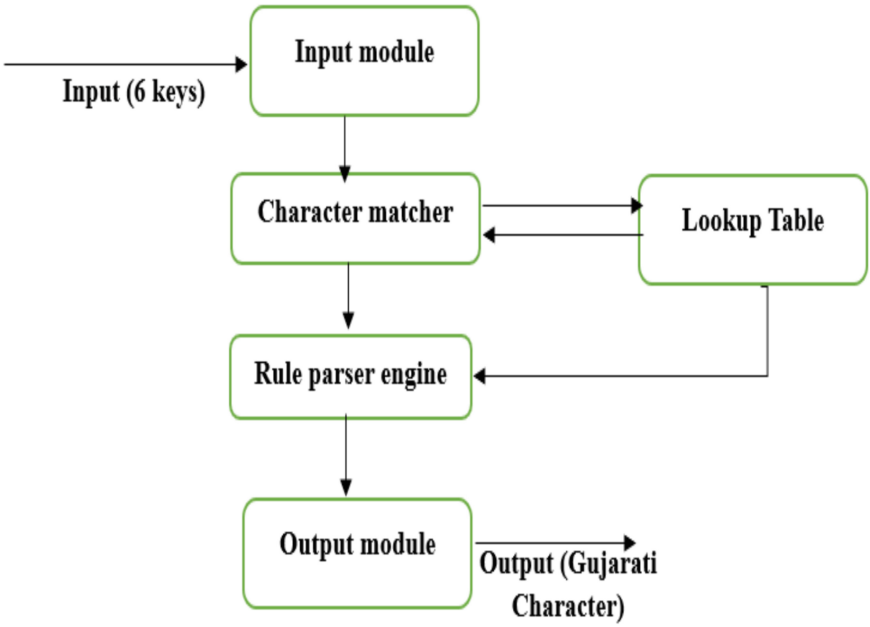


Fig. 10. Block diagram of Proposed Transliteration Model

3.1 Phases of the Proposed Transliteration Model

Input module – the input module is further divided into two components. Input panel and input processor. Input panel will accept the input in terms of 6 keys (SDFJKL) and further pass on those keys to the Input Processor to generate Braille cells. These 6 keys represent 6 Braille dots within the cell. The keys are gathered from the user and then converted into the Braille pattern (cell).
Lookup table – act as a collection of Braille cells with its corresponding Gujarati language character used for character mapping.
Character matcher – at the time of cross-script transliteration the transliteration is done from one script to another. So, to transliterate from source script i.e. in this research Braille to destination script i.e. Gujarati character mapping technique is used.

It tries to match the Braille pattern with the collection present in the lookup table and retrieve the equivalent Gujarati character with the help of character mapping technique.
Rule parser engine – Both the source and destination scripts have their own syntax and rules. In the proposed model, taking this into consideration during transliteration rules and syntax of both the scripts are parsed and according to that characters are mapped.
Output module – the Gujarati character that is generated after applying the rules and parsing it, is displayed to the user.

3.2 Transliteration Rules

Following rules are taken into consideration by the proposed transliteration model [30]:

Through Braille cells only 64 unique patterns can be generated. But Gujarati language set of 75 characters. There are some characters in Gujarati Braille which are similar and indistinguishable. So, there is a need to take assumptions and applying some indicators.

For example, in Gujarati Braille ૧ is represented in the similar way as ચ, ૨ is represented in the similar way as ખ and so on. So, number indicator will be the extra Braille cell present in Braille script at the time of the representation of the numerals in Braille script.

There are no distinct characters present in Gujarat Braille to indicate compound or half or conjunct characters. So, here as well additional character known as half character indicator is used to indicate that the following character is the half or compound or character.

So, at the time of typing Braille visually impaired people will consider all those extra indicators in their document. But in Gujarati script all those indicators are not available.

During transliteration from Braille to Gujarati all of those indicators meaning are to be considered and then it needs to be transliterated into the corresponding text.

3.3 Algorithm

An algorithm is developed to transliterate Bhartiya Braille characters in plain Gujarati characters that can further be printed on the printer to be read by sighted people.

The steps are as follows:

Step 1: Fetch the keys from the keyboard that is pressed by the user (SDF JKL).

Step 2: The keys are combined to formed the pattern of Braille cell (six dots) in difference of one second.

F $\rightarrow$ 1st dot	J $\rightarrow$ 4th dot
D $\rightarrow$ 2nd dot	K $\rightarrow$ 5th dot
S $\rightarrow$ 3rd dot	L $\rightarrow$ 6th dot

Step 3: The character matcher of the model matches the pattern of Braille cell dots with the entries within the lookup table.

Step 4: If match is found for the pattern of dots of the Braille cell, then the resultant Gujarati character is parse against the rules (as mentioned in the 3.2 transliteration rules) created in the rule parser engine.

Step 5: The formed Gujarati character is displayed (as shown in Fig. 11 of topic 3.4).

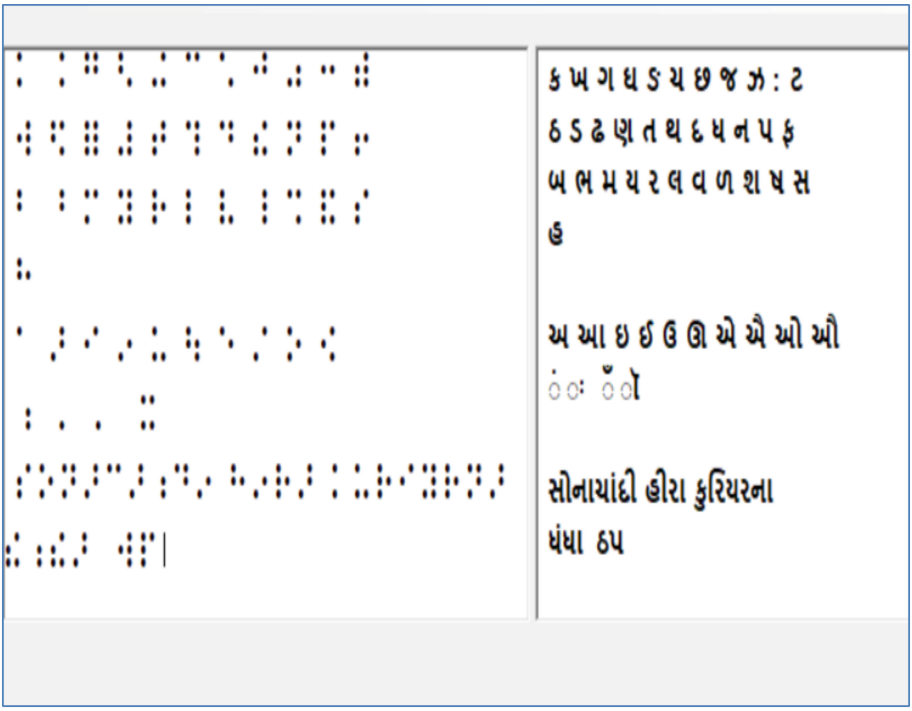


Fig. 11. Transliteration of Braille character into Gujarati Text through the proposed model

3.4 Experimental Results

The model is tested by typing Braille characters i.e. ⠁ (અ) to ⠒ (ઃ) and ⠅ (ક) to ⠷ (ઙ) and words that doesn't contains compound and conjunct characters. A sample transliteration done by the model is shown in the Fig. 11. The Gujarati language text generated through the model is verified by both the Braille expert as well as Gujarati language teacher. They did not found error in the transliteration of whole character and words without compound and conjunct characters. So, it shows that model works perfect for plain Gujarati Braille characters and gives satisfactory result.

4 Conclusion

The paper explains the fundamentals of Gujarati language and Braille writing. Authors have also discussed the need of the research by providing the statistics as well as the challenges that are faced by visually impaired people at the time of written communication. Additionally, it describes that some researchers have worked on the Braille character recognition in several languages such as Arabic, English, Hindi, Bengali etc. Reasonable amount of work is found in Gujarati Braille recognition but complete transliteration into Gujarati language is still untouched. Therefore, the researcher believed that the Braille to

Gujarati text transcription was necessary. Proposed transliteration model in this article will convert Braille cells inputted by the user into Gujarati language characters. The proposed work is tested and it shows satisfactory result. So, it will act as an aid for blind and visually challenged people of the society.

5 Future Enhancement

In the present model recognition and transliteration of the individual Braille cells into Gujarati characters is carried out. In future, the model will be extended to transliterate Braille numeral, words as well as sentences that contain compound and conjunct characters in to Gujarati language characters by considering all the Braille syntax and the rules.

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